43. Which one of the following is the pathway to increase productivity in agriculture?

(a) Efficient irrigation (b) Quality seeds

(c) Use of pesticides (d) Use of fertilizers

(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (e)

All the statements given above will increase productivity in agriculture.

44. What is/are true in reference to Agriculture Census (2015-16) ?

(i) Small and marginal farmers are 86.2% of total farmers whereas they hold only 47.3% of farm land (out of total farm land)

(ii) Statewise, based on total number of farmers, Maharashtra is at third position

(iii) Statewise, based on total farmed land area, Rajasthan is at first position.

(a) (i), (ii) and (iii) (b) (i) and (iii)

(c) (i) and (ii) (d) Only (i)

Chhattisgarh P.C.S. (Pre) 2021

Ans. (a)

According to the Agriculture Census (2015-16), small and marginal farmers constitute 86.2 percent of the total farmers, while they own only 47.3 percent of the cropped areas. On the basis of state-wise total cultivable land, Rajasthan rank first. On the basis of state-wise total number of farmers, Maharashtra is at the third position.

Green Revolution

***Norman Ernest Borlaug** is the father of Green Revolution in the world. He was born on 25th March 1914 in **Iowa** State of America.

Wheat production has been doubled in India-Pakistan and Mexico due to a high yielding variety of seeds developed by him. Borlaug is among those seven persons in the world who got all the three awards viz. **Nobel Prize**, 'Presidential Medal of Freedom' and **Congressional Gold Medal**.

*Borlaug has been honoured with **Padma-Vibhusan** of India in 2006. *Father of the Green Revolution in India is **Dr. M.S. Swaminathan**. Under his leadership, High Yielding Varieties of wheat has been successfully developed in India and the use of High Yielding variety of seeds started in India. Apart from this, special emphasis was given on the increased use of fertilizers and adequate development of irrigation facilities. The programme for the use of developed seeds in India was started by the cooperation of Rockefeller Foundation of U.S.A.

*First stage of Green Revolution was carried out in India from **1966** to **1981.** Its second stage was implemented in the period of **1981-1995**. Its third stage was started in the year 1995 and covered almost all the areas of the country. *The most benefitted crop in production and productivity by the green revolution in India is wheat.

Before Green Revolution, the production of wheat all over India was 12.3 Million tons whereas in 2000-2001 it became 69.68 million tons and in 2023-24 is 113.3 million tons.

The increase occurred due to the increase in yield per Hectare. After wheat, the highest impact of the Green Revolution was seen on rice cultivation. *The main plant in the Green Revolution was the wheat of Mexican species which was brought from **Borlaug's International Maize and Wheat Improvement Center (CIMMYT).**

In the year, 2023-24 the production of wheat reached up to 3559 kg/Hectare. ***High yielding variety of seeds, fertilizer**, **irrigation**, **use of technology**, **marketing**, and **industrialization of agriculture** is known as Green Revolution.

On 28 July 2000, the Central Government launched a new National Agricultural Policy. Under this policy the concept of rainbow revolution was proposed. In this policy, all the revolutions directly or indirectly related to agricultural activities were included.

Various Revolutions and its Areas	
Green Revolution	Food production or crop production
Golden Revolution	Fruit and Vegetable production
White Revolution	Milk Production
Brown Revolution	Fertilizer Production / Non-conventional Energy
Blue Revolution	Fish Production
Red Revolution	Meat/Tomato Production
Black Revolution	Petroleum Production
Round Revolution	Potato Production
Yellow Revolution	Oilseed Production
Pink Revolution	Onion Production/Export of Meat

'National Zero till seed cum fertilizer drill' developed by G.B. Pant University of Agriculture and Technology. This machine enables to sow directly after paddy harvest without prior seed-bed preparation.

1. **Green Revolution was the result of adaptation of the new Agricultural Strategy, which was introduced in the 20th century during decades of –**

(a) Fifties (b) Seventies
(c) Sixties (d) Eighties

U.P.P.C.S. (Pre) 2015

Ans. (b)

Green Revolution started in the 7th decade in the year 1966 which was its first stage and The continued until till 1981. In the first stage, Haryana, Punjab and Western Uttar Pradesh were included.

2. **The term 'Evergreen Revolution' has been used for increasing agricultural production in India by –**

(a) Norman Barlogue (b) M.S. Swaminathan
(c) Raj Krishna (d) R.K.V. Rao

U.P.P.C.S. (Mains) 2015

Ans. (b)

The term 'Evergreen Revolution' was used for increasing agricultural production in India. It was used by Father of India's Green Revolution, Dr. M.S. Swaminathan. He described 'Evergreen Revolution' as increasing productivity in perpetuity without ecological harm. Dr. Swaminathan has proposed the following steps for the second Green Revolution.

(i) Use of bio-fertilizers and compost along with chemical fertilizers.
(ii) Rainwater harvesting land development to facilitate like industries.
(iii) Emphasis on agricultural economic zones and contract farming.

3. **Norman Ernest Borlaug who is regarded as the father of the Green Revolution is from which country?**

(a) United States of America
(b) Mexico
(c) Australia
(d) New Zealand

I.A.S. (Pre) 2008

Ans. (a)

Norman Ernest Borlaug was born on 25 March 1914 in Cresco, Iowa, U.S.A. He took up an agricultural research position in Mexico, where he developed semi-dwarf, high-yield, disease-resistant wheat varieties. During the mid-20th century, Borlaug led the introduction of these high-yielding varieties combined with modern agricultural production techniques to Mexico, Pakistan, and India. He is one of seven people to have won the Nobel Peace Prize, the Presidential Medal of Freedom and the Congressional Gold Medal. He was also awarded the Padma Vibhushan, India's second highest civilian honor in 2006.

4. **The 'Father of the Green Revolution' in the world is**

(a) Norman E. Borlaug (b) M. S. Swaminathan
(c) G. S. Khush (d) B. P. Pal

U.P. Lower Sub. (Pre) 2015

Ans. (a)

See the explanation of the above question.

5. **Who among the following was closely associated with the Green Revolution?**

(a) Dr. Swaminathan (b) Kr. Kurien
(c) C. Subrahmaniam (d) Dr. A.P.J. Abdul Kalam

M.P.P.C.S. (Pre) 1999
Chhattisgarh P.C.S. (Pre) 2003

Ans. (a)

Dr. M.S. Swaminathan is known as the "Father of India's Green Revolution" for his leadership and success in introducing and further developing high-yielding varieties of wheat in India. He was closely related to the Green Revolution in India.

6. **Green Revolution means –**

(a) Use of green manure
(b) Grow more crops
(c) High yield variety programme
(d) Green vegetation

R.A.S./R.T.S.(Pre) 1999

Ans. (c)

Green Revolution means using high-yielding varieties of seeds, modifying farm equipment, and substantially increasing chemical fertilizers. In, India Green Revolution began in 1966-67. High yield varieties programme started in India with the help of the Rockefeller Foundation based in the U.S.A.

7. **Which one of the following most appropriately describes the nature of the Green Revolution of the late sixties of 20th century?**

(a) Intensive cultivation of green vegetable
(b) Intensive agriculture district programme
(c) High-yielding varieties programme
(d) Seed-Fertilizer-Water technology

64th B.P.S.C. (Pre) 2018

Ans. (c)

See the explanation of the above question.

8. **Green Revolution is related to**

(a) Millet production (b) Pulse production
(c) Wheat production (d) Oilseed production

U.P.P.C.S. (Pre) 2015

Ans. (c)

The Green Revolution in India was started in 1966-67 which led to an increase in food grain production. The main development was higher yielding varieties of wheat which increased the production to 2.5 times. Its credit is given to American Agronomist Dr. Norman Borlaug and M.S. Swaminathan of India.

9. Which one of the following crops is the highest beneficiary of the Green Revolution in both production and productivity:

(a) Jawar (b) Maize
(c) Rice (d) Wheat

U.P.P.C.S.(Pre) 1994
U.P.P.C.S.(Pre) 2001

Ans. (d)

Wheat is the highest beneficiary of the Green Revolution in both production and productivity followed by rice.

10. The impact of the Green Revolution was felt most in the production of

(a) Oilseed (b) Wheat
(c) Sugarcane (d) Pulses

U.P.P.C.S. (Mains) 2004
U.P.P.C.S. (Pre) 1990

Ans. (b)

See the explanation of the above question.

11. After Independence India progressed maximum-

(a) In the production of Rice
(b) In the production of Pulses
(c) In the production of Jute
(d) In the production of Wheat

40th B.P.S.C. (Pre) 1995

Ans. (d)

India achieved maximum growth in the production of wheat after Independence. This growth was propelled by the Green Revolution introduced in 1966-67.

12. Which was the main crop used in the Green Revolution?

(a) Japonica rice (b) Indian rice
(c) Amer wheat (d) Mexican wheat

Uttarakhand P.C.S. (Pre) 2010

Ans. (d)

In the decade of 1960, the introduction of high yielding varieties of seeds and the increased use of chemical fertilizers and irrigation is termed the Green Revolution in India. The credit for Green Revolution is given to Norman Borlaug (in Global context) and M.S. Swaminathan (in the Indian context). The main crop used in Green Revolution was Mexican Wheat recieved from Borlaug's International Maize and Wheat Improvement Center based in Mexico.

13. As a result of `Green Revolution' the yield per hectare of wheat touched the record figure of :

(a) 1500 kg (b) 2000 kg
(c) 2222 kg (d) 3000 kg

U.P.P.C.S. (Mains) 2003

Ans. (b)

As a result of the Green Revolution, the yield per hectare of wheat touched a record figure of 2000 Kg (three times as earlier). 2023-24 Yield of wheat crops is 3559 kg/hectare.

14. What is true about the second Green Revolution in India?

1. It aims at further increasing the production of wheat and rice in areas already benefited from the green revolution.
2. It aims at extending seed-water-fertilizer technology to areas which hitherto could not benefit from green revolution.
3. It aims at increasing yields of crops other than those used for the green revolution in the beginning.
4. It aims at integrating cropping with animal husbandry, social forestry and fishing

Select the correct answer from the code given below :

(a) 1 and 2 (b) 2 and 3
(c) 2 and 4 (d) 1 and 4

47th B.P.S.C. (Pre) 2005

Ans. (c)

The Second Green Revolution in India aims at extending seed, water, fertilizer and technology to areas which could not be benefited from the Green Revolution and integrating agriculture with animal husbandry, social forestry and fishing. It does not aim to yield the crop other than used in the Green Revolution in the beginning. Thus, statement II and IV are correct.

15. Select the component of the Green Revolution by using the given code :

1. High-yielding varieties of seeds
2. Irrigation
3. Rural Electrification
4. Rural roads and marketing

Code :

(a) Only (1) and (2) (b) Only (1),(2) and (3)
(c) Only (1),(2) and (4) (d) All four

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (d)

In 1965, the introduction of high-yielding varieties of seeds (hybrid seeds), increased use of chemical fertilizers, irrigation, rural electrification and connectivity to the market places through roads led to the increase in production needed to make the country self-sufficient in food grains which was called the Green Revolution. Thus all the four components were part of the Green Revolution.

16. Assertion (A) : Green Revolution has resulted in the growth of food grain production in India.

Reason (R) : Regional disparities have been aggravated due the Green Revolution in India.

Code :

(a) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(b) (A) is false, and (R) is true.
(c) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(d) (A) is true, and (R) is false.

R.A.S./R.T.S. (Pre) 2016

Ans. (a)

The Green Revolution in India was a period when agriculture in India resulted in the growth of food grain production. The adoption of new agricultural technology, high yielding varieties of seeds, plant breeding, irrigation etc. increased the production of food grain. Thus Assertion (A) is correct. Green Revolution spread only in irrigated and high potential rain fed areas. The villages or regions without the access to sufficient water were left out. Thus it led to regional disparities. Thus, Reason (R) is correct but (R) is not the correct explanation of (A).

17. Rainbow Revolution is related with –

(a) Green-Revolution (b) White-Revolution
(c) Blue-Revolution (d) All the above

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (d)

On 28 July 2000, the new National Agricultural Policy was announced by the Central government. In this policy the concept of Rainbow Revolution was introduced. The various colours of the Rainbow Revolution indicate various farm practices such as Green Revolution (Foodgrains), White Revolution (Milk), Yellow Revolution (Oilseeds), Blue Revolution (Fisheries), Red Revolution (Tomatoes),Golden Revolution (Fruits), Grey Revolution (Fertilizers) and so on. Thus, the concept of the Rainbow Revolution is an integrated development of crop cultivation, horticulture, forestry, fishery, poultry, animal husbandry and food processing industry.

18. Match List-I with List-II and select the correct answer from the codes given below :

List – I	**List – II**
A. Growth in Food Production	**1. Green Revolution**
B. Milk Production	**2. Blue Revolution**
C. Fisheries	**3. White Revolution**
D. Fertilizers	**4. Grey Revolution**

Code :

	A	B	C	D
(a)	1	3	2	4
(b)	3	1	4	2
(c)	2	4	3	1
(d)	3	2	4	1

U.P.P.C.S. (Pre) 2014

Ans. (a)

The correct match between list-I and list-II is as follows:

Growth in food Productions	- Green Revolution
Milk Production	- White Revolution
Fisheries	- Blue Revolution
Fertilizers	- Grey Revolution

19. Pink Revolution is associated with –

(a) Cotton (b) Garlic
(c) Grapes (d) Onion

U.P.P.C.S. (Mains) 2010

Ans. (d)

Pink Revolution is associated with Onion.
Other important Revolutions are as follows –
Black Revolution - Petroleum.
Red Revolution - Tomato and Meat
Blue Revolution - Fisheries
Yellow Revolution - Oilseed

20. The Black Revolution is related to the

(a) Fish production (b) Coal production
(c) Crude oil production (d) Mustard production
(e) None of the above/More than one of the above

60th 62nd B.P.S.C. (Pre) 2016

Ans. (c)

See the explanation of the above question.

21. Match List-I with List-II and select the correct answer from the codes given below :

List – I (Revolution)	**List – II (Related with)**
A. Golden Revolution	**1. Oil Seed Production**
B. Grey Revolution	**2. Horticulture and Honey**
C. Yellow Revolution	**3. Petroleum Production**
D. Black Revolution	**4. Fertilizers**

Code :

	A	B	C	D
(a)	2	3	4	1
(b)	4	2	1	3
(c)	2	4	1	3
(d)	1	2	3	4

U.P.P.C.S. (Pre) 2022

Ans. (c)

The correct match between list-I and list-II is as follows :

Golden Revolution	–	Horticulture and Honey
Grey Revolution	–	Fertilizers
Yellow Revolution	–	Oil Seed Production
Black Revolution	–	Petroleum Production

22. Which of the following is not correctly matched?

Revolution	**Related to**
(a) Golden	Horticulture
(b) White	Milk
(c) Blue	Poultry
(d) Green	Agriculture

U.P.P.C.S. (Mains) 2016

Ans. (c)

The Blue Revolution refers to the remarkable emergence of aquaculture. Aquaculture refers to all form of active culturing of aquatic animals, plants and fishes. It is not related to poultry. Thus, it is not correctly matched. All other options are correctly matched.

23. The 'Blue Revolution' is associated with

(a) Agriculture (b) Iron and Steel Industry
(c) Irrigation (d) Fishing

Uttarakhand P.C.S. (Mains) 2006

Ans. (d)

See the explanation of the above question.

24. Which of the following is associated with "Blue Revolution"?

(a) Animal (b) Bee
(c) Fisheries (d) Poultry

U.P.U.D.A./L.D.A. (Spl) (Mains) 2010

Ans. (c)

See the explanation of the above question.

25. Among the following, which one is related to the Blue Revolution in India?

(a) Floriculture (b) Sericulture
(c) Pisciculture (d) Horticulture

Chhattisgarh P.C.S. (Pre) 2016
M.P.P.C.S. (Pre) 2016

Ans. (c)

Blue Revolution in India is related to Pisciculture. Pisciculture involves raising fish commercially in enclosures or tank. The breeding, rearing and transplantation of fish by artificial means are called Pisciculture. India's Blue Revolution focused on fish production.

26. Among the following which one is related to Blue Revolution in India?

(a) Horticulture (b) Floriculture
(c) Pisciculture (d) Sericulture
(e) None of these

Chhattisgarh P.C.S. (Pre) 2016

Ans. (c)

Blue Revolution in India is related to Pisciculture.

27. "Blue Revolution" is related with the following

(a) Food grain production
(b) Oilseed production
(c) Fish production
(d) Milk production

U.P. R.O./A.R.O. (Pre) (Re-Exam) 2016

Ans. (c)

See the explanation of the above question.

28. Zero Till Seed cum-Fertilizer Drill was developed at

(a) P.A.U. Ludhiana
(b) G.B. Pant University of Agriculture and Technology, Pantnagar
(c) IISR, Lucknow
(d) IARI, New Delhi

U.P. U.D.A./L.D.A. (Spl) (Pre) 2010

Ans. (b)

"Zero Till Seed cum-Fertilizer Drill" was developed by G.B. Pant University of Agriculture and Technology, Pant Nagar. This machine enables sowing directly after paddy harvest without prior seedbed preparation. This saves the diesel, tractor's working time, labour and most importantly it gives higher yield.

29. Which one of the following is not correctly matched with regard to the revolution in agriculture ?

(a) White : Milk (b) Green : Food grains
(c) Golden : Horticulture (d) Blue : Poultry

U.P.P.C.S. (Mains) 2013

Ans. (d)

Blue Revolution is related to fisheries not poultry. Thus option (d) is not correctly matched.